Experiment No : 2

Date of Performance:
Date of Submission:

Aim: Simulation of basic communication system (Tx–Channel–Rx) using MATLAB / Python / online tools.

Required Equipment / Software :

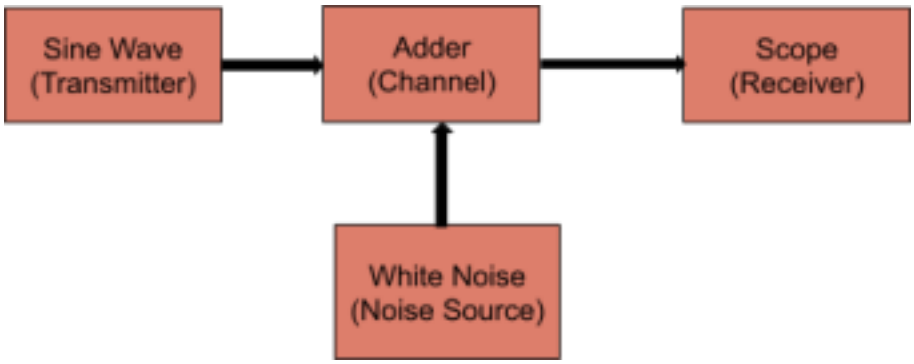
Hardware: PC / Laptop with internet access Software / Online Resources: Web browser to access PathView Simulator.

Theory:

A wireless communication system transmits information from a source to a destination using electromagnetic waves as the transmission medium. The three fundamental building blocks of any communication system are the transmitter, the channel, and the receiver. In a typical communication system, the information source generates a message (voice, data, or video). The transmitter processes this message into a form suitable for transmission over the channel. The channel is the physical medium through which the signal travels — in wireless systems, this is the atmosphere or free space. During transmission, the signal is corrupted by noise and interference. The receiver captures the signal from the channel, processes it, and attempts to recover the original information.

a) Transmitter (Tx): The transmitter converts the information signal into a form suitable for transmission. In this experiment, we use a sine wave generator as the transmitter. **b) Wireless Channel:** The channel is the medium through which the signal travels from the transmitter to the receiver. In wireless communication, the channel is the atmosphere. The transmitted signal is affected by various impairments, including: • Path Loss: Signal strength decreases with distance. • Noise: Unwanted random signals that corrupt the transmitted signal (e.g., thermal noise, atmospheric noise). • Interference: Signals from other wireless systems operating on the same frequency. In this experiment, we simulate the channel by adding noise to the transmitted signal. **c) Receiver (Rx):** The receiver captures the signal from the channel and processes it. In this experiment, we use a Scope block to visualize the received signal.

Experimental Setup : Since this is a software-based simulation experiment, the setup involves accessing the PathView website and building the simulation using drag-and-drop blocks.



Procedure

1. Open your browser, go to view.pathsim.org and press Enter.

2. Wait for the website to load completely. You will see a blank canvas (the workspace) with a sidebar on the left containing the Node Library marked as **'Blocks'**.

3. Important: No sign-up, login, or installation is required. The simulator runs entirely in your browser.

4. Wait for the website to load completely. You will see a blank canvas (the workspace) with a sidebar on the left containing the **Blocks** library.

5. **Add the Transmitter (Signal Source)**:

- Click on the 'Blocks' panel on the left side of the screen.
- Search for the 'SinusoidalSource'. This block generates a sine wave signal.
- Click on it. A small block 'SinusoidalSource' will appear on the canvas.

6. **Add the Noise Source (Channel Impairment):**

- In the **'Blocks'** panel, search for the **'WhiteNoise'** block.
- This block generates white noise, which we will use to simulate channel noise.
- Click on it. A small block 'WhiteNoise' will appear on the canvas.

7. **Add the Adder (Combine Signal and Noise):**

- In the **'Blocks'** panel, search for the **'Adder'** block.
- This block adds two signals together.
- Click on it. A small block 'Adder' will appear on the canvas.

8. **Add the Receiver (Signal Visualizer):**

- In the 'Blocks' panel, search for the 'Scope' block.
- This block displays the signal as a waveform (like an oscilloscope). ● Click on it. A small block 'Scope will appear on the canvas. Your canvas should now have four blocks: SinusoidalSource, WhiteNoise, Adder, and Scope.

9. **Connect the Transmitter to the Adder:**

- Move your mouse over the SinusoidalSource block. You will see a small circle (port) on its right side. This is the output port. The mouse pointer (cursor) will change from hand icon to + (plus) sign.
- Click and hold the output port of the SinusoidalSource block.
- Drag your mouse to the first input port of the Adder block (on its left side). A wire will appear as you drag.
- Release the mouse button to complete the connection.

10. **Connect the Noise Source to the Adder:**

- Click and hold on the output port (right side) of the WhiteNoise block. o Drag

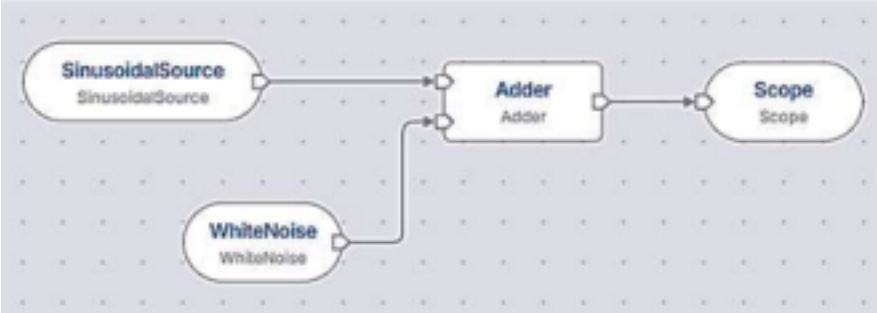
your mouse to the second input port of the Adder block (on its left side).

- Release the mouse button to complete the connection.

11. **Connect the Adder to the Receiver:**

- Click and hold on the output port (right side) of the Adder block. ● Drag your mouse to the input port of the Scope block (on its left side). ● Release the mouse button to complete the connection.

Your block diagram should now look like this:



12. Look for a ‘Run’ button in the top toolbar of the PathView interface. It may be a triangular "play" icon (▶).

13. Click the ‘Run’ button. The simulation will start.

14. A plot window will open, showing the received signal (sine wave with noise). Adjusting the Noise Level

15. Double click the WhiteNoise block on the canvas, a ‘Properties’ panel will pop-up.
16. Find the standard_deviation parameter. This controls the strength (amplitude) of the noise. o Its default value is 1.0.

17. To increase the noise: Change the value to 2.0 or 5.0.

18. To decrease the noise: Change the value to 0.5 or 0.1.

19. To remove noise entirely: Change the value to 0.0.

20. After changing the value, click the ‘Run’ button again to see the new result. Observe how the waveform on the Scope changes as you increase or decrease the noise level. Data Recording

21. For each noise level setting (standard_deviation), observe the signal on the Scope.

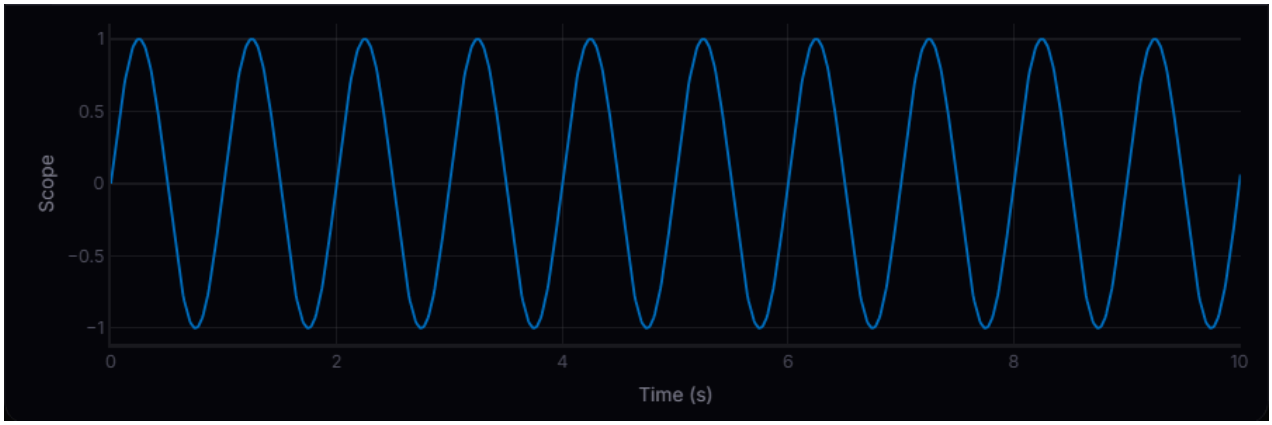
22. Record your observations in the table below.

23. Note how the signal quality changes as the noise level increases or decreases.

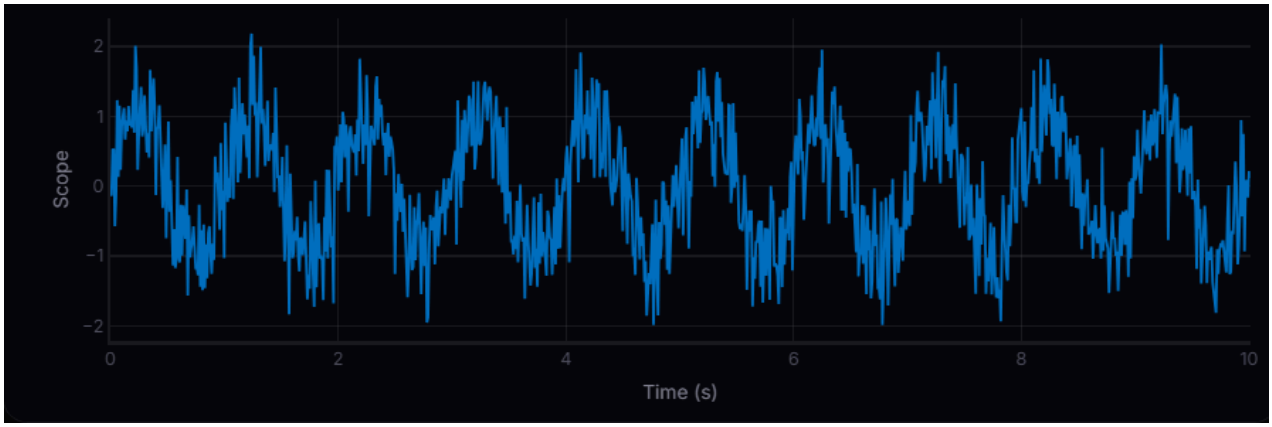
Observation Table A : Effect of Noise Level on Signal Quality

Noise Level (standard_deviation)	Signal Quality (Clean / Slightly Noisy / Very Noisy / Almost Lost)	Description of Observed Waveform
0.0	Clean	Continuous Waveform
0.1	Slightly Noisy	Small Ripples
0.5	Slightly Noisy	Visible Distortion
2.0	Very Noisy	Masked Peak
5.0	Almost Lost	Pure Noise

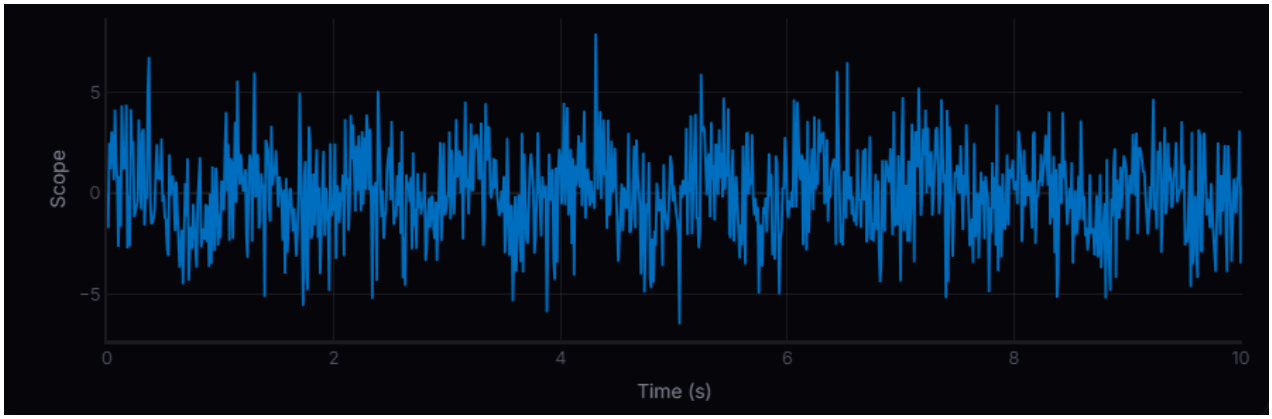
Noise Level = 0 (i.e. transmitted signal without noise)



Noise Level = 0.5



Noise Level = 2.0.



Observation Table B : Comparison of Transmitted and Received Signal

Parameter	Transmitted Signal (Sine)	Received Signal (Sine + Noise)
Waveform Shape	Smooth sine	Distorted Sine
Amplitude	Constant	fluctuating
Noise Presence	No noise	High Noise
Signal Quality	Perfect	Degraded

Analysis and Discussion :

1. From Table 1, observe the relationship between noise level and signal quality. What general trend do you observe?

As the noise level increases, the signal quality degrades inversely. At low noise levels, the waveform remains recognizable with minimal distortion, but at higher noise levels, the original signal is increasingly masked until it becomes lost.

2. At a very high noise level (e.g., standard_deviation = 5.0), what do you observe about the received signal? Is it still recognizable as a sine wave?

At a very high noise level (standard_deviation = 5.0), the received signal becomes completely distorted and appears as random noise. The noise amplitude heavily dominates over the signal amplitude, masking the underlying structure. No, it is no longer

recognizable as a sine wave.

3. At very low noise level (e.g., standard_deviation = 0.1), what do you observe about the received signal? How does it compare to the original sine wave?

At a very low noise level (standard_deviation = 0.1), the received signal closely resembles the original sine wave, with only slight high-frequency ripples visible along its path. The general shape, frequency, and amplitude of the sine wave remain clearly preserved.

Result :

The simulation of a basic communication system (Tx-Channel-Rx) was successfully performed using PathView. The effect of channel white noise on a transmitted sinusoidal signal was observed and analyzed across various noise levels.

Conclusion:

In a basic communication system, the transmission channel introduces white noise, which directly degrades signal quality. As the noise standard deviation increases, signal distortion increases and amplitude fluctuates, causing the received waveform to transition from clean to completely lost. Low noise levels allow the sine wave to remain recognizable, whereas high noise levels render it unrecoverable without proper filtering.

Viva Voce Questions :

1. What is a wireless channel? What impairments does it introduce?
2. What is noise? How does it affect signal quality?
3. What is the effect of increasing noise on the received signal?

MARKS & SIGNATURE :

R1 (3 Marks)	R2 (5 Marks)	R3 (4 Marks)	R4 (3 Marks)	Total (15 Marks)	Signature